



**Laboratorium
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Laporan Akhir Praktikum Jaringan Komputer

Vlan, Trunk, OSPF, dan MultiVendor

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1 Langkah-Langkah Percobaan

Untuk praktikum ini, pastikan sudah membuat topologi terlebih dahulu sebelum mengikuti langkah-langkah percobaannya.

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch (`enable → configure terminal`).
2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing). Verifikasi menggunakan perintah `show vlan brief`.
3. Mengonfigurasi *access port* pada interface Switch yang mengarah ke PC client sesuai dengan alokasi VLAN masing-masing.
4. Mengonfigurasi *trunk port* pada interface Switch (Gi0/0) yang mengarah ke Cisco Router dan mengizinkan VLAN 10, 20, 30, dan 40 lewat. Verifikasi menggunakan `show interfaces trunk`.
5. Menyimpan konfigurasi dengan perintah `write memory`.

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Masuk ke terminal Cisco Router (`enable → configure terminal`).
2. Menghidupkan interface fisik (Gig0/2) yang terhubung ke Switch tanpa menggunakan IP address (`no ip address` dan `no shutdown`).
3. Mengonfigurasi *subinterface* (Gig0/2.10, Gig0/2.20, dst.) menggunakan enkapsulasi *dot1Q* dan memberikan IP Address gateway untuk masing-masing VLAN.
4. Mengonfigurasi DHCP-Server pada Cisco Router untuk memberikan IP otomatis kepada client di VLAN 10 dan VLAN 20.
5. Memberikan IP Static secara manual pada PC (VPCS) yang berada di VLAN 30 dan VLAN 40.
6. Memverifikasi IP pada Cisco Router (`show ip interface brief`, `show ip dhcp binding`) dan melakukan uji *ping* ke gateway maupun antar-VLAN dari PC client.

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengonfigurasi IP address jaringan LAN (192.168.50.1/24) pada Huawei Router dan mengatur IP VPC pada LAN Huawei.
2. Mengonfigurasi IP address untuk Link 1 (10.10.10.0/30) dan Link 2 (10.10.11.0/30) antara Cisco Router dan Huawei Router.
3. Mengonfigurasi IP address link antara Huawei Router dan MikroTik (10.10.20.0/30).
4. Mengonfigurasi IP address link antara Cisco Router dan MikroTik (10.10.30.0/30).
5. Memverifikasi status interface tiap router dan melakukan *ping* ke router tetangga terdekat (*directly connected*) untuk memastikan konektivitas fisik berjalan baik.

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengonfigurasi OSPF pada Cisco Router dengan mendaftarkan seluruh *network* lokal dan mengiklankannya ke Area 0.
2. Mengonfigurasi OSPF pada Huawei Router untuk Area 0 dan mendaftarkan *network* yang terhubung dengannya.
3. Mengonfigurasi OSPF pada MikroTik dengan menambahkan *network* penghubung ke dalam area *backbone*.
4. Mengatur nilai *OSPF Cost* pada masing-masing interface di tiap router. Cost = 10 untuk dua jalur utama (Cisco ↔ Huawei), dan Cost = 100 untuk jalur *backup* (melewati MikroTik).
5. Memverifikasi relasi *neighbor* OSPF dan *routing table* untuk memastikan setiap perangkat telah menerima rute secara dinamis dari router lain.
6. Menguji *ping end-to-end* dari PC VLAN Cisco menuju PC LAN Huawei dan sebaliknya.

1.5 Praktikum 5: Pengujian Failover OSPF

1. Melakukan pengecekan jalur normal (dua link Cisco-Huawei aktif) menggunakan perintah `trace 192.168.50.10` dari PC VLAN. Pastikan trafik melewati jalur utama.
2. Mematikan salah satu jalur utama (`shutdown` interface Gi0/3 pada Cisco) dan melakukan `trace` ulang untuk melihat bahwa trafik tetap melewati link utama yang tersisa.
3. Mematikan jalur utama yang kedua (`shutdown` interface Gi0/0 pada Cisco) dan melakukan `trace` ulang. Pastikan trafik *failover* dan berpindah melewati jalur *backup* (MikroTik).
4. Menghidupkan kembali kedua interface link utama pada Cisco dan memverifikasi bahwa jalur penyaluran kembali normal ke rute utama.

2 Analisis Hasil Percobaan

Setelah membuat topologi dan mengikuti alur dari modul, dapat disimpulkan bahwa praktikum yang pertama hingga praktikum ke lima bisa sukses dijalankan.

3 Hasil Tugas Modul

Tugas modul tersedia di Github. **KLIK DISINI!**

4 Kesimpulan

Berdasarkan serangkaian percobaan dari Praktikum 1 hingga Praktikum 5, dapat disimpulkan bahwa praktikan telah berhasil menyimulasikan dan mengonfigurasi infrastruktur jaringan *enterprise* berskala menengah hingga besar menggunakan perangkat *multivendor* (Cisco, Huawei, dan MikroTik). Pada tahap awal, praktikan berhasil memecah jaringan fisik menjadi beberapa segmen logis (VLAN 10,

20, 30, dan 40) menggunakan Cisco Switch, serta mengimplementasikan konsep *Router-on-a-Stick* (ROAS) dan DHCP Server pada Cisco Router agar setiap VLAN dapat saling berkomunikasi secara terpusat dan efisien. Selanjutnya, perbedaan arsitektur vendor tidak menjadi hambatan berkat penerapan protokol standar terbuka OSPF (*Open Shortest Path First*). Ketiga *router* tersebut terbukti mampu membentuk relasi *neighbor* dan bertukar informasi *routing table* secara dinamis, sehingga memungkinkan komunikasi *end-to-end* dari segmen Cisco ke segmen LAN Huawei. Di samping itu, praktikan juga berhasil melakukan skenario manajemen jalur berbasis *OSPF Cost*, di mana jalur langsung Cisco–Huawei ditetapkan sebagai rute utama (cost 10) dan rute via MikroTik sebagai cadangan (cost 100).

5 Lampiran

5.1 Hasil Screenshot Praktikum 1: Konfigurasi Cisco Switch

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	FINANCE	active	
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
Switch#
```

Gambar 1: Hasil verifikasi VLAN pada Cisco Switch

```
*Jun  6 03:33:49.589: %SYS-5-CONFIG_I: Configured from console
Switch#
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1

```
Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
Switch#
```

Gambar 2: Hasil verifikasi Trunking pada Cisco Switch

5.2 Hasil Screenshot Praktikum 2: Konfigurasi Cisco Router

```
*Jun 6 03:42:42.862: %SYS-5-CONFIG_I: Configured from console by c
Router#
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  up                    up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up                    up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3       unassigned      YES unset  administratively down  down
Router#
Router#
```

Gambar 3: Verifikasi IP Interface pada Cisco Router

```
Router#
Router#show ip dhcp pool

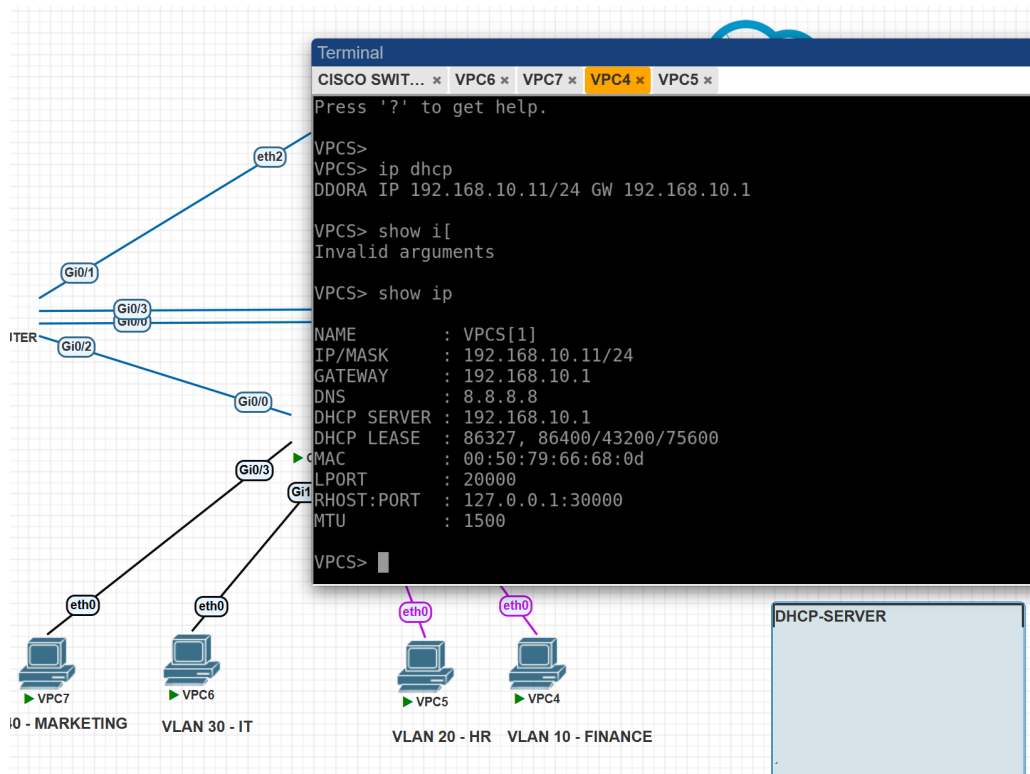
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#
Router#
```

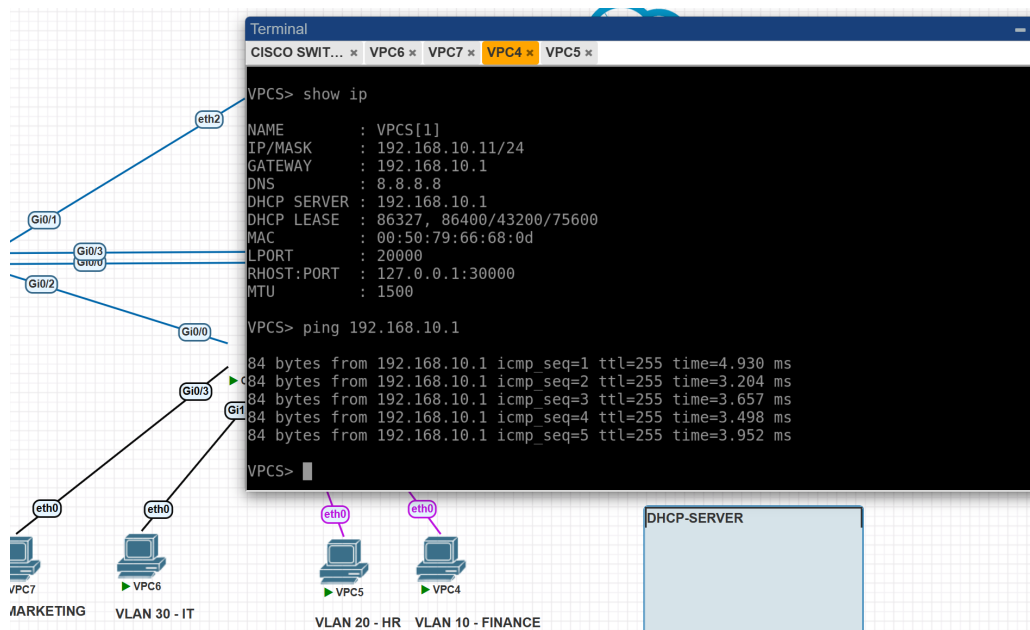
Gambar 4: Verifikasi DHCP Pool pada Cisco Router

```
Current index      IP address range      Leased addresses
192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/             Lease expiration        Type
                   Hardware address/
                   User name
192.168.10.11        0100.5079.6668.0d      Jun 07 2026 03:42 AM    Automatic
192.168.20.11        0100.5079.6668.0e      Jun 07 2026 03:42 AM    Automatic
Router#
Router#
```

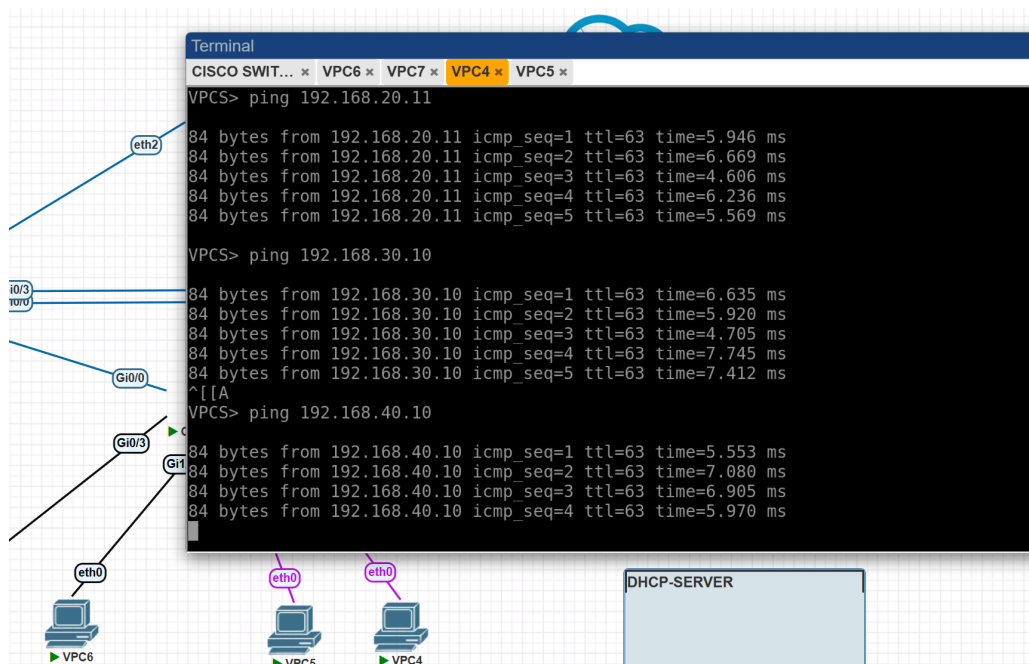
Gambar 5: Verifikasi DHCP Binding pada Cisco Router



Gambar 6: Verifikasi IP Client (DHCP) pada PC



Gambar 7: Uji Ping ke Gateway



Gambar 8: Uji Ping Antar-VLAN

5.3 Hasil Screenshot Praktikum 3: Konfigurasi Huawei Router dan IP Address Antar-router

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.434 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=5.378 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.991 ms
^C
VPCS>
```

Gambar 9: Verifikasi LAN Huawei

```
Router#
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0                      10.10.11.1      YES manual up                    up
GigabitEthernet0/1                      10.10.30.1      YES manual up                    up
GigabitEthernet0/2                      unassigned      YES unset  up                    up
GigabitEthernet0/2.10                   192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20                   192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30                   192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40                   192.168.40.1    YES manual up                    up
GigabitEthernet0/3                      10.10.10.1      YES manual up                    up
Router#
Router#
```

Gambar 10: Verifikasi Interface pada Cisco Router

```
CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  MIKROTIK  x
Error: Incomplete command found at '^' position.
[~HUAWEI]
[~HUAWEI]quit
<HUAWEI>
<HUAWEI>display ip interface brief
*down: administratively down
!down: FIB overload down
*down: standby
(L): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                                IP Address/Mask  Physical  Protocol VPN
Ethernet1/0/0                            10.10.10.2/30    up        up      --
Ethernet1/0/0.4094                       128.1.138.137/16 up        up      l3vpn
Ethernet1/0/1                            unassigned       up        down    --
Ethernet1/0/1.4094                       128.1.138.137/16 up        up      l3vpn
Ethernet1/0/2                            192.168.50.1/24 up        up      --
Ethernet1/0/2.4094                       128.1.138.137/16 up        up      l3vpn
Ethernet1/0/3                            10.10.20.1/30    up        up      --
Ethernet1/0/3.4094                       128.1.138.137/16 up        up      l3vpn
Ethernet1/0/4                            10.10.11.2/30    up        up      --
Ethernet1/0/4.4094                       128.1.138.137/16 up        up      l3vpn
Ethernet1/0/5                            unassigned       up        down    --
---- More ----
```

Gambar 11: Verifikasi Interface pada Huawei Router

```
repeat new password>
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >
```

Gambar 12: Verifikasi Interface pada MikroTik

```
GigabitEthernet0/3 10.10.10.1 VLS manual up up
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/7 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/3 ms
Router#
```

Gambar 13: Uji Ping dari Cisco Router

```
<HUAWEI>
<HUAWEI>ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

<HUAWEI>
<HUAWEI>
```

Gambar 14: Uji Ping dari Huawei Router

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                          56 255 4ms
1 10.10.20.1                          56 255 1ms
2 10.10.20.1                          56 255 1ms
3 10.10.20.1                          56 255 1ms
4 10.10.20.1                          56 255 1ms
5 10.10.20.1                          56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                          56 255 2ms
1 10.10.30.1                          56 255 1ms
sent=2 received=2 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > |
```

Gambar 15: Uji Ping dari MikroTik

5.4 Hasil Screenshot Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

```
Gateway of last resort is not set

Router#
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0

Router#
```

Gambar 16: Verifikasi Network OSPF Cisco

```
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
2.2.2.2        1     FULL/BDR        00:00:36   10.10.11.2    GigabitEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:38   10.10.10.2    GigabitEthernet0/3
Router#
Router#
```

Gambar 17: Verifikasi OSPF Neighbor pada Cisco

```
CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  MIKROTIK  x  +  v  -

[~HUAWEI]commit
[~HUAWEI]
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.10.1         BDR: 10.10.10.2      MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.11.1         BDR: None          MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

Gambar 18: Verifikasi OSPF Peer pada Huawei

```
[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=10s
[admin@MikroTik] > |
```

Gambar 19: Verifikasi OSPF Neighbor pada MikroTik

```
CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  MIKROTIK  x  +
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O        10.10.20.0/30 [110/110] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
          [110/110] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
O        192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
          [110/11] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
Router#
Router#
```

Gambar 20: Verifikasi Tabel Routing OSPF pada Cisco

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface
-----
10.10.30.0/30       OSPF     10     110      D  10.10.11.1             Ethernet1/0/4
10.10.30.0/30       OSPF     10     110      D  10.10.10.1             Ethernet1/0/0
192.168.10.0/24      OSPF     10     11        D  10.10.11.1             Ethernet1/0/4
192.168.20.0/24      OSPF     10     11        D  10.10.10.1             Ethernet1/0/0
192.168.30.0/24      OSPF     10     11        D  10.10.11.1             Ethernet1/0/4
192.168.30.0/24      OSPF     10     11        D  10.10.10.1             Ethernet1/0/0
192.168.40.0/24      OSPF     10     11        D  10.10.11.1             Ethernet1/0/4
192.168.40.0/24      OSPF     10     11        D  10.10.10.1             Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface
-----
10.10.10.0/30       OSPF     10     10        10.10.10.2             Ethernet1/0/0
10.10.11.0/30       OSPF     10     10        10.10.11.2             Ethernet1/0/4
10.10.20.0/30       OSPF     10     100       10.10.20.1             Ethernet1/0/3
192.168.50.0/24     OSPF     10     1         192.168.50.1           Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
```

Gambar 21: Verifikasi Tabel Routing OSPF pada Huawei

```

[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0   Ado  10.10.10.0/30      10.10.20.1    110
    10.10.30.1
1   Ado  10.10.11.0/30      10.10.20.1    110
    10.10.30.1
2   Ado  192.168.10.0/24     10.10.30.1    110
3   Ado  192.168.20.0/24     10.10.30.1    110
4   Ado  192.168.30.0/24     10.10.30.1    110
5   Ado  192.168.40.0/24     10.10.30.1    110
6   Ado  192.168.50.0/24     10.10.20.1    110
[admin@MikroTik] >

```

Gambar 22: Verifikasi Tabel Routing OSPF pada MikroTik

```

VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS>

```

Gambar 23: Pengujian End-to-End: Ping ke LAN Huawei

```

CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PC  x  HUAWEI ROUTER  x
OSPF  10  11  D  10.10.10.1  Ethernet1/
OSPF routing table status : <Inactive>
Destinations : 4  Routes : 4
Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
10.10.10.0/30  OSPF  10  10  10.10.10.2  Ethernet1/
10.10.11.0/30  OSPF  10  10  10.10.11.2  Ethernet1/
10.10.20.0/30  OSPF  10  100  10.10.20.1  Ethernet1/
192.168.50.0/24  OSPF  10  1  192.168.50.1  Ethernet1/
<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>ping 192.168.10.11
PING 192.168.10.11: 56 data bytes, press CTRL_C to break
  Reply from 192.168.10.11: bytes=56 Sequence=1 ttl=63 time=4 ms
  Reply from 192.168.10.11: bytes=56 Sequence=2 ttl=63 time=4 ms
  Reply from 192.168.10.11: bytes=56 Sequence=3 ttl=63 time=4 ms
  Reply from 192.168.10.11: bytes=56 Sequence=4 ttl=63 time=5 ms
  Reply from 192.168.10.11: bytes=56 Sequence=5 ttl=63 time=4 ms

--- 192.168.10.11 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 4/4/5 ms
<HUAWEI>
<HUAWEI>

```

Gambar 24: Pengujian End-to-End: Ping dari LAN Huawei

```
VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 6.215 ms 3.868 ms 3.852 ms
 2 10.10.11.2 5.338 ms 5.235 ms 5.901 ms
 3 *192.168.50.10 8.304 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.614 ms 3.631 ms 4.510 ms
 2 10.10.30.2 4.652 ms 3.447 ms 3.378 ms
 3 10.10.20.1 11.464 ms 6.592 ms 5.997 ms
 4 *192.168.50.10 6.087 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.942 ms 5.270 ms 4.586 ms
 2 10.10.11.2 7.938 ms 5.845 ms 5.319 ms
 3 *192.168.50.10 10.986 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

```
Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:14:14 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:14:14 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router(config)#
Router(config)#
```